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*Department of Electrical, Electronic and Computer Engineering*

## EPE 321 - Group Design Document

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# 1 Task

As part of the project component of EPE 321, each group is required to submit a group design document detailing each aspect of the project selected for the group. Each group must produce and submit one document containing the finalised design for implementing the project.

The group design document must be clear and concise, highlighting the following about the design:

- As an introduction, a section must be included indicating how the work will be divided between each team member. Also, each team member's associated responsibilities must be noted. This detail must also be included on the team's Git repository and will be approved by the lecturers.
- UML must be used as a notation in the design of artefacts.
- The final design must incorporate a class diagram of the project, indicating relationships between different classes.
- An activity diagram must be provided that details the project's functionality at a high level. This model must incorporate the best-case scenarios and edge cases, or problem states, that the projects must accommodate. A textual description of the flow must accompany each activity diagram.
- For each class embedded in the design, provide the unit tests required to test the class adequately. Unit tests should be shown in a table with the following columns: number (e.g. test 5), tested component/class, inputs and expected output.
- Develop and document a high-level GUI design through example figures or screenshots (a mock design showing the layout and placement of elements). Any tool can be used to develop the design.
- A product and sprint backlog should be included as screenshots from Jira. Backlogs will help with the planning and execution of the project, and perfect estimation is not expected here. We expect to see 50% of the tasks completed by the progress demo, with the remainder completed by the individual demo. The group can decide if they will have only one sprint until the progress demo or if they will have multiple shorter sprints.
- The due date for this assignment is 23:59 on **September 09, 2025**<sup>1</sup>. Submit a **PDF** version of the document to **both** the Turnitin system via the link on ClickUP and the AMS. One submission per group is sufficient.

A portion of your marks for this document will be for formatting, grammar, and language, so keep this in mind. Be sure to include a plagiarism statement. Take note of the AI Chatbot/generative tools policy in the EPE study guide.

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<sup>1</sup>See the EECE General Study Guide 8.4.2 for details on the procedure for submission times.

## 2 Notes

- You should re-use the selected project's UML class diagrams produced in the SRS for this assignment (just be sure to update them to lower-level detail where necessary). This is your final design, and it needs to have the lowest design level of granularity to avoid implementation and integration issues. The lowest level refers to the following: all variables have types in a diagram, and functions show their return values and parameter lists.
- Groups will gain access to their repositories before this document is handed in. The member responsibilities must also be stated in the README of your repository and will thus be pre-approved before this document is submitted. Please ensure your responsibilities are indicated by 1 September 2025 in your repository. Your group will receive an email if there are any issues.
- You are recommended to keep your group design under 20 pages. Only if you need to use more may you do so. Note that designs showing the same information clearly but in as few pages as necessary tend to be more elegant.
- There is no template provided. You are welcome to structure your document to make the most sense to your group.

### 3 Marking rubric

|                                               |                                          |                                                                                                     |                                                                                                                               |                                                                                                                    |
|-----------------------------------------------|------------------------------------------|-----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>Formatting, grammar and spelling (3)</b>   | Lacks serious effort, several errors.    | Three or fewer errors.                                                                              | Two or fewer errors                                                                                                           | One or No errors. Excellent document.                                                                              |
| <b>Overview and team responsibilities (2)</b> | Incomplete/missing/lacks detail.         | Some effort, but the design will likely be hard to change into code.                                | The design is clear enough for minimal issues to be encountered                                                               | High-detail, unambiguous and exceptional explanations for design. Also not too drawn out on singular aspects       |
| <b>Backlogs (5)</b>                           | Incomplete/missing/lacks detail.         | Some effort was made, poor planning, too vague, lacks detail.                                       | Nearly complete, lacks attention to detail/not complete enough for good time planning.                                        | Very clear and clearly shows time management, set goals and proper breakdown of tasks per sprint, for each sprint. |
| <b>Unit tests tables (5)</b>                  | Incomplete/missing/lacks detail.         | Major gaps in system testing, not sufficient testing to ensure quality.                             | Clear/ comprehensive unit tests with some minor gaps with vagueness/uncertainty. Not testing for every possible critical case | Comprehensive unit tests with little to No room for error. Quality software should be guaranteed.                  |
| <b>UML class diagrams (5)</b>                 | Not given/Too high level/confusing/vague | Some aspects were clear, but most were not. Lacks too much detail / too high level for final design | Class diagrams are good, but lacks small aspects like relationships and interfaces                                            | Clear and concise design that can easily translate into code with minimal issues.                                  |
| <b>UML activity diagrams (5)</b>              | Not given/Too high level/confusing/vague | Some aspects were clear, but most were not. Lacks too much detail / too high level for final design | Flow is good but lacks alternate or error conditions.                                                                         | Clear and concise design that can easily translate into code with minimal issues. Edge cases catered for.          |
| <b>High-level GUI design. (5)</b>             | Not given/vague.                         | Some aspects were clear, but most were not.                                                         | Most elements were presented.                                                                                                 | Clear and concise mock design that showed all elements.                                                            |